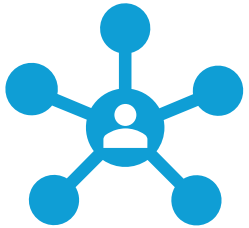
RETIREMENT READINESS

Helping Users Understand the Plan for Retirement

Problem We Are Solving

Retirement concepts are confusing (401k, Roth IRA, etc.)

People don’t know how to plan or where to start



Financial information online is scattered and hard to trust. Users need simple, reliable explanations

Our solution

AI-powered app that answers retirement questions

Provides clear explanation in simple language

Uses real financial sources (Fidelity, IRS, etc.)

What the app does

User asks a retirement question

System generates an AI response

Answers are grounded and traceable to its source

Displays answer + sources + grounding report

```
Loaded 43 PDF definition chunks.
Loaded 32 Fidelity chunks from https://www.fidelity.com/learning-center/smart-money/what-is-a-roth-ira
Loaded 38 Fidelity chunks from https://www.fidelity.com/learning-center/smart-money/what-is-an-ira
Loaded 29 Fidelity chunks from https://www.fidelity.com/learning-center/smart-money/what-is-a-401k
Loaded 28 Fidelity chunks from https://www.fidelity.com/retirement-ira/401k-rollover-ira
Loaded 9 Fidelity chunks from https://www.fidelity.com/learning-center/smart-money/what-is-a-roth-401k
Loaded 30 Fidelity chunks from https://www.fidelity.com/learning-center/trading-investing/compound-interest
Loaded 7 IRS chunks from https://www.irs.gov/retirement-plans/roth-iras
Loaded 9 IRS chunks from https://www.irs.gov/retirement-plans/individual-retirement-arrangements-iras
Loaded 7 IRS chunks from https://www.irs.gov/retirement-plans/401k-plans
```

Retrieval Strategy

- **Frontend:**
  - User input is sent and retrieved in the backend
- **Backend:**
  - The backend it a topic-based retrieval system
  - This keyword is used to retrieve its respective document with chunks of information
  - Then Gemini is given a prompt and is asked to use only the information it was given
- Ensures the model only sees only verified text

Validator & Repair Loop

- **Validator:**
  - Checks if the model's answer is grounded, cited correctly, and uses only numbers from the retrieved chunks
- **Repair Loop:**
  - If invalid, the answer is rewritten using only the retrieved chunks
  - Up to 3 repair attempts before the system refuses
- Prevents hallucinations and keeps answers simple, correct, and source-based

Grounding Strategy

- **Backend:**
  - Each key phrase (3-4 words) is checked and flags any unsupported claim
  - Unsupported claims will trigger a repair loop
- **Frontend:**
  - The UI shows a **Grounding Report**

Grounding Report (6)

most basic Individual Retirement: Supported

- ▶ *Chunk Northwestern Mutual_definitions_1* — Northwestern Mutual (definitions)
- ▶ *Chunk Northwestern Mutual_definitions_4* — Northwestern Mutual (definitions)

deductible contributions: Supported

- ▶ *Chunk Northwestern Mutual_definitions_4* — Northwestern Mutual (definitions)
- ▶ *Chunk Northwestern Mutual_definitions_5* — Northwestern Mutual (definitions)

The deductibility of contributions: Not supported

modified adjusted gross income: Supported

- ▶ *Chunk Northwestern Mutual_definitions_3* — Northwestern Mutual (definitions)
- ▶ *Chunk Northwestern Mutual_definitions_5* — Northwestern Mutual (definitions)

sponsored retirement: Supported

- ▶ *Chunk Northwestern Mutual_definitions_1* — Northwestern Mutual (definitions)
- ▶ *Chunk Northwestern Mutual_definitions_2* — Northwestern Mutual (definitions)
- ▶ *Chunk Northwestern Mutual_definitions_3* — Northwestern Mutual (definitions)
- ▶ *Chunk Northwestern Mutual_definitions_5* — Northwestern Mutual (definitions)

contributions: Supported

- ▶ *Chunk Northwestern Mutual_definitions_2* — Northwestern Mutual (definitions)
- ▶ *Chunk Northwestern Mutual_definitions_4* — Northwestern Mutual (definitions)
- ▶ *Chunk Northwestern Mutual_definitions_5* — Northwestern Mutual (definitions)

Joaquin Castillo (Backend & AI)

- Built core backend and core frontend
- Handled backend to frontend data flow and vice versa
- Implemented grounding + validation
- Created segmented response + citations formatting

Javier Castillo (Backend & Data)

- Developed chunking and generator logic
- Improved evaluation test cases and documentation
- Worked on repair loops
- Enhanced backend performance

Abcde Mireles (Backend & UI)

- Created PRD documentation & UI design
- Implemented refusal mechanism & test system
- Expanded data sources & Improved chunking
- Verified scenario engine & Built demo script and notes

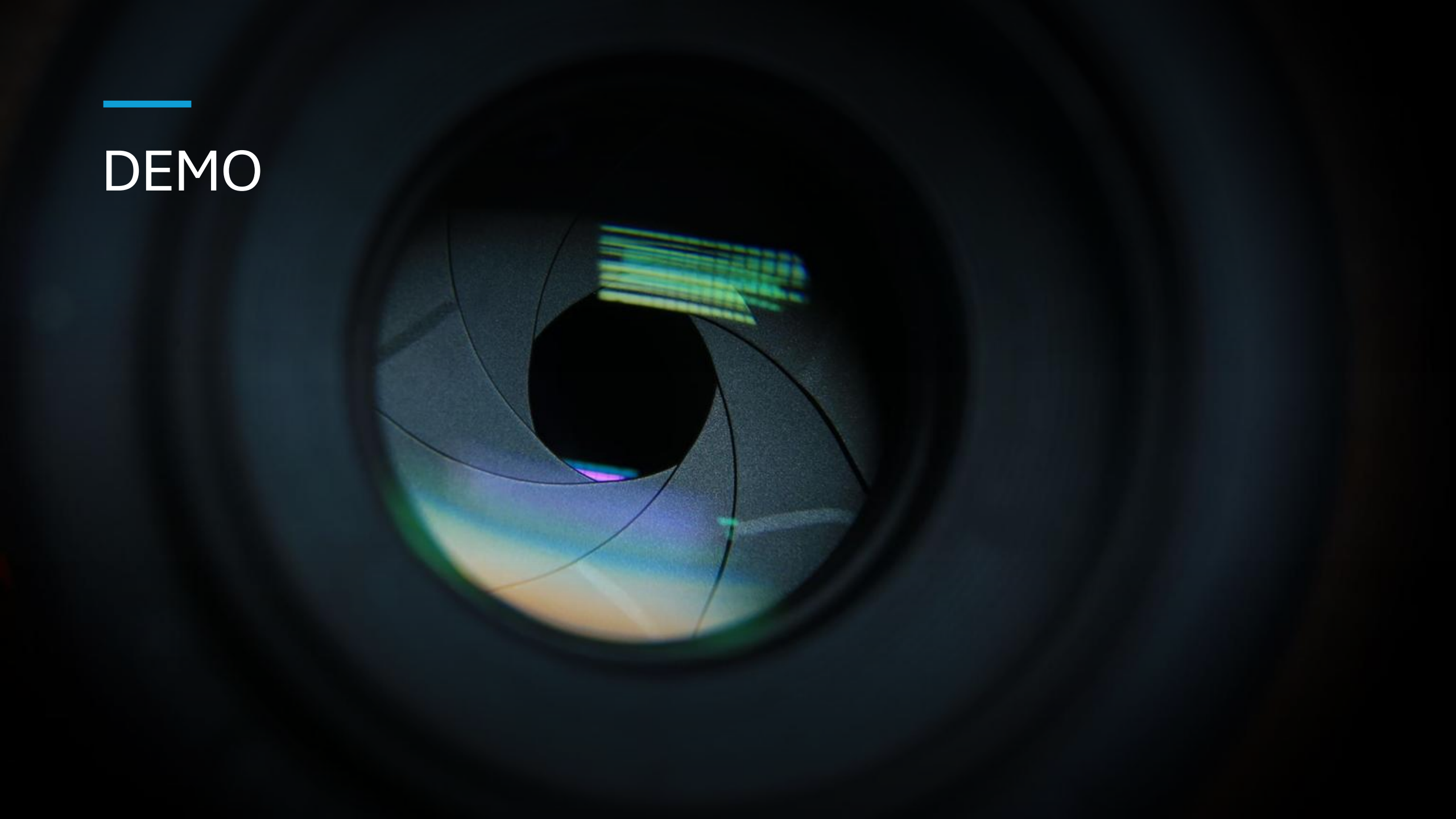
Jose Torres (Integration & Testing)

- Implemented fallback AI system
- Fixed backend/frontend integration
- Updated citations and validation logic
- Handled repo cleanup and environment set up

Retirement Readiness Evaluation Results Sprint 3

Retrieval Accuracy	Citation Coverage	Grounding Accuracy	Hallucination Rate	Refusal Accuracy
94%	100%	80.6%	0%	94%

DEMO



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Mathematical Verification For Scenario

	A	B
1	Age	25
2	Retirement Age	65
3	Annual Income	$ 60,000.00
4	Current Saving	$ 1.00
5	Monthly Contribution	$ 500.00
6	Return Rate	3.50%
7	Years to Grow	40
8	Monthly Rate	0.2917%
9	Total Months	480
10	Furture Value of current savings	$ 4.05
11	Future Value of Contributions	$522,333.44
12	Total Projected Balance	**$522,337.49**

Questions?